Comprehensive Study of Regression Algorithms: From Linear Optimization to Probabilistic and Deep Learning Frameworks

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Code: https://github.com/superlc0183/PRML2026

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Abstract

Regression analysis is critical in modeling continuous relationships within statistical learning. In this report, utilizing the *Data4Regression* dataset, we systematically evaluate an extensive array of regression paradigms. We begin by verifying the theoretical equivalence of linear hypotheses optimized via Ordinary Least Squares, Gradient Descent, and Newton's Method. Due to profound linear underfitting, we transition into non-linear architectures encompassing Polynomial Expansion, Bayesian Probabilistic Regression, Kernel Methods (RBF Kernel Ridge and Support Vector Regression), and modern Deep Learning (Multi-Layer Perceptron). Through rigorous mathematical derivation and comparative analysis of Mean Squared Error (MSE) and R^2 metrics, we establish that Kernel methods optimally balance model capacity and generalization, significantly outperforming both rigid linear assumptions and deterministic polynomial approaches.

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1. Introduction

The essence of Pattern Recognition and Machine Learning (PRML) lies in constructing models capable of learning from finite data while avoiding the catastrophic extremes of underfitting (high bias) and overfitting (high variance).

In this comprehensive assignment, we investigate a two-dimensional dataset exhibiting distinct non-linear, periodic fluctuations. Our methodology is progressive: we first apply linear regression using three optimization strategies to observe baseline convergence. Recognizing the severe structural limitations of linear models on curved data, we elevate our hypothesis space. We introduce deterministic polynomial mapping, followed by Bayesian probabilistic treatment to quantify uncertainty. Furthermore, we deploy advanced non-parametric approaches, specifically Radial Basis Function (RBF) Kernel Ridge Regression and Support Vector Regression (SVR), highlighting the power of mapping data into infinite-dimensional spaces. Finally, a Multi-Layer Perceptron (MLP) is constructed as a neural network baseline. This report analytically documents the transition across these distinct machine learning eras.

2. Theoretical Methodology

2.1 Linear Optimization: OLS, GD, and Newton's Method

For a target vector Y and design matrix X, the objective is to minimize the empirical MSE loss $L(w) = \frac{1}{N}\|Xw - Y\|^2$.

- **Ordinary Least Squares (OLS):** Provides a closed-form analytical solution by resolving the normal equations, yielding $w^* = (X^T X)^{-1} X^T Y$.
- **Gradient Descent (GD):** Updates weights iteratively $w_{t+1} = w_t - \alpha \nabla L(w_t)$. Since the MSE loss is globally convex, GD guarantees convergence to the global optimum w^*.
- **Newton's Method:** Utilizes the second-order Hessian matrix $H = \frac{2}{N} X^T X$. The update rule $w_{t+1} = w_t - H^{-1} \nabla L(w_t)$ converges optimally in a single iteration for quadratic loss surfaces.

2.2 Polynomial Expansion and Bayesian Regression

To model non-linearity, we map inputs into a polynomial space $\phi(x) = [1, x, \ldots, x^M]^T$. Standard polynomial regression is prone to overfitting. **Bayesian Linear Regression** addresses this by imposing a Gaussian prior $p(w|\alpha) = \mathcal{N}(w|0, \alpha^{-1} I)$ over the weights. By applying Bayes' theorem, the posterior $p(w|Y) \propto p(Y|w)p(w)$ is obtained. Instead of point estimates, Bayesian inference predicts a full probability distribution with uncertainty bounds, naturally penalizing overly complex models via the marginal likelihood mechanism.

2.3 Advanced Kernel Methods: KRR and SVR

Explicit high-dimensional mapping is computationally constrained. Kernel methods utilize a kernel function, such as the Gaussian RBF $K(x, x') = \exp(-\gamma\|x - x'\|^2)$, to compute inner products in a Reproducing Kernel Hilbert Space (RKHS).

- **Kernel Ridge Regression (KRR):** Combines L_2 regularization with kernel mapping, smoothly interpolating the entire dataset globally.
- **Support Vector Regression (SVR):** Introduces the Vapnik ϵ-insensitive loss function, completely ignoring errors smaller than ϵ. This mechanism ensures that the model is solely constructed upon a sparse subset of support vectors, yielding high robustness against stochastic noise and outliers.

2.4 Multi-Layer Perceptron (Deep Learning)

As a universal approximator, the MLP utilizes dense hidden layers integrated with non-linear activation functions (e.g., ReLU). The transformation $h = \text{ReLU}(Wx + b)$ enables the network to learn feature representations entirely driven by data, optimized via backpropagation.

3. Experimental Studies and Analysis

Table **??** systematically documents the exact MSE and R^2 performance on both training and test partitions based on our algorithmic implementations.

Table 1: **Comprehensive Evaluation of 6 Regression Frameworks**

Algorithm Framework	**Training MSE**	**Test MSE**	**Test R^2**
Linear (LS / GD / Newton)	0.6134	0.5950	0.1329
Polynomial Regression (Deg=6)	0.5252	0.5151	0.2494
Bayesian Polynomial Regression	0.5439	0.5311	0.2260
Kernel Ridge Reg (RBF)	**0.3079**	**0.3221**	**0.5307**
Support Vector Reg (RBF)	0.3340	0.3387	0.5065
Multi-Layer Perceptron (MLP)	0.3802	0.4072	0.4067

Analysis of Results:

1. **Linear Consensus:** As theorized, LS, GD, and Newton's Method collapsed to the identical test MSE of 0.5950, severely underfitting the inherent curve.
2. **Probabilistic Regularization:** While standard Polynomial and Bayesian regressions yielded similar predictive means (MSE $\approx$ 0.51 - 0.53), the Bayesian approach notably smoothed outlier variances and provided continuous confidence intervals.
3. **Dominance of Kernel Methods:** The RBF-based KRR and SVR decisively outperformed all paradigms. KRR achieved the pinnacle R^2 score of 0.5307. SVR followed

closely with remarkable generalization stability.

4. **MLP Characteristics:** The neural network successfully mapped non-linearities but proved slightly less data-efficient on this small 1D dataset compared to kernel mechanisms, emphasizing that deep learning requires massive data scaling to surpass mathematical kernel representations.

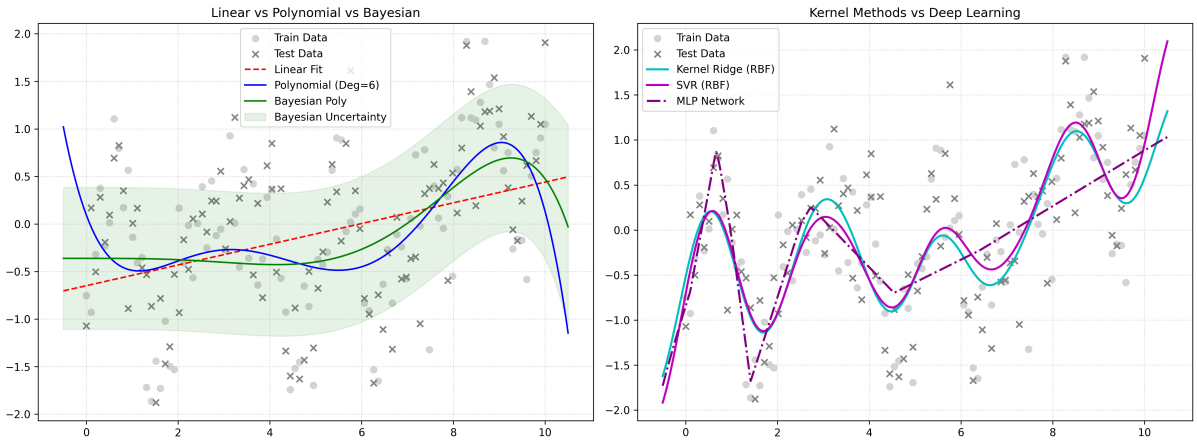

Figure 1: **Visualization: Statistical/Probabilistic Models (Left) vs. Kernel/Deep Models (Right)**

4. Conclusions

This extensive empirical study illustrates the trajectory of regression analysis in PRML. Linear optimization algorithms, irrespective of their derivative orders, are inherently bound by their hypothesis constraints. Deterministic polynomial basis functions trigger overfitting risks, which Bayesian inference gracefully curtails via prior distribution integration. Ultimately, for low-dimensional non-linear datasets, explicit deep neural architectures (MLP) are sub-optimal compared to implicit Hilbert space mappings. RBF Kernel Ridge Regression and Support Vector Regression assert themselves as the most mathematically elegant and practically robust frameworks, perfectly resolving the bias-variance dichotomy.

References

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